

When Is the Optimum Radiological Response to Proton Beam Therapy in Uveal Melanoma?

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Keywords

Uveal melanoma · Proton beam therapy · Radiological response

Abstract

Introduction: Proton beam therapy (PBT) is an effective treatment option for uveal melanomas. Following treatment, it may take many months for the tumour to respond and it may initially enlarge. We reviewed our PBT patients to determine when they showed a radiological response to treatment. **Methods:** Patients undergoing PBT for ciliary body or choroidal melanomas between 2008 and 2018 were included. Data were collected on patient demographics, treatments before and after PBT and survival. All ultrasound investigations prior and since PBT were reviewed and tumour volume calculated using a validated formula for a half-ellipsoid shape. **Results:** 193 patients were analysed, 169 with choroidal and 24 with ciliary body melanomas. 31.6% patients had other treatment prior to PBT. At a mean of 8 months post-PBT, 64.7% of patients had a reduced tumour volume with 20.2% having larger tumours. At a mean of 15 months post-treatment, these figures were 67.8% and 10.3%. 18.1% of patients had an enucleation during the

study period. The earliest responses were seen at 2 months, the latest at 32 months post-treatment. 5-year melanoma-specific survival was 82.3%. **Conclusions:** In our study, by 6 months post-PBT, a majority of patients show a reduction in tumour volume. Of those that do not, many respond in the next 6 months and a response may be seen up to 32 months after treatment. Patients may need to be monitored for up to 32 months to see a final response to PBT treatment.

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Introduction

Uveal melanomas are the commonest form of primary intraocular malignancy in adults. They typically present as a pigmented mass in the choroid (90%), ciliary body (6%), and iris (4%). There is no significant gender predilection, but it is more common in Caucasians. Uveal melanoma is rare, with an incidence of around 6 per million population per year in Europe. Patients with uveal melanoma typically present with blurred or altered vision, or asymptotically after routine ophthalmological examination [1–3]. Uveal melanoma characteristically metastasises via haematogenous spread to the liver. The

15-year melanoma-specific survival in Scotland is 83.8%, 81.3%, and 100% for choroidal, ciliary body, and iris melanomas, respectively [3]. Fifteen-year survival for in Sweden for choroidal and iris melanomas has been reported as 84% and 92%, respectively [4], and 5- and 10-year melanoma-specific survival figures for choroidal melanoma in the USA have been reported as 87% and 78.5% [5], respectively.

Proton beam therapy (PBT) is one of the modalities used to treat uveal melanomas. It can be used as a primary treatment, as a salvage treatment in tumour recurrence, or as a neoadjuvant treatment prior to surgical resection. The main aims of PBT are to eliminate the threat of metastatic disease whilst preserving the eye and any meaningful vision possible [6].

PBT utilises ionising radiation in the form of charged, high-energy protons. These high-energy protons are released from an atomic nucleus by means of a particle accelerator. The protons do not scatter inside the cellular nuclei on arrival to their target tumour cell. This feature is due to the protons relatively large size. This allows a very focused beam of radiation to reach the target cells with minimal effect on the tissue surrounding the tumour. Another important trait of PBT is the Bragg peak phenomenon. This allows for the delivery of a high dose of radiation to a very localised site, thus improving the effectiveness of treatment whilst, once again, minimising radiation exposure to the surrounding tissues. On arrival to the target tumour cells, the high-energy protons will quickly deposit most of their energy here, thereby effectively eliminating any exit dose of radiation. The protons damage the reproductive ability of the tumour cells by disrupting DNA, which in turn leads the tumour cells to enter senescence or to undergo apoptosis [7–9].

At our centre, PBT is typically used as a primary treatment when the tumour is too large for ruthenium plaque brachytherapy and for small, posterior tumours to limit radiation to the optic disc and fovea [10]. Patients treated with PBT have a similar survival to those treated with enucleation. Risks include local tumour recurrence, vision loss secondary to collateral damage via radiation papillopathy or retinopathy, macular oedema, retinal detachment, or neovascular glaucoma [6, 11]. Risk of enucleation post-PBT at 5 years is approximately 5.4–14% with the two commonest reasons being local tumour recurrence and neovascular glaucoma [6]. In a Scottish cohort, 22.4% of patients had an enucleation post-PBT with 78.8% of these occurring in the first 3 years following treatment. 48% of these were for recurrence and neovascular glaucoma in 42% [12]. No large-scale studies exist looking at the change in tumour size radiologically

following PBT, but a smaller study suggested a decrease in tumour height occurred on average 12 months from treatment but ranged between 1 and 24 months [13].

We studied radiological response of uveal melanomas to PBT in Scotland. The primary aim of our study was to determine when PBT patients for both ciliary body and choroidal melanoma showed a radiological response to treatment.

Methods

We undertook a retrospective cohort study. All patients seen at the Scottish Ocular Oncology Service (SOOS), Tennent Institute of Ophthalmology, Glasgow between April 2008 and April 2018 and referred for PBT at the Clatterbridge Cancer Centre NHS Foundation Trust were identified from the Ocular Oncology Service database. Those patients who did not have proton beam treatment for a diagnosis of choroidal, ciliary body, or iris melanoma were excluded. Data from each patient's electronic health record (Clinical Portal, Orion Health, Auckland, New Zealand) on diagnosis, laterality, prior treatments for melanoma, presence or absence of metastases, and survival were collected.

Ocular ultrasound scans were accessed via the board's PACS imaging system (Carestream Vue, Carestream Health, Rochester, NY, USA). All scans were performed or supervised by one of two radiologists who specialise in ocular ultrasound (O.C. or W.K.). The scan just prior to PBT was taken as the initial scan for comparison purposes. This was usually performed within 1 month of the start of PBT. We used a validated formula for calculation of an ellipsoid volume in the context of choroidal melanoma survival $(4/3\pi a^2 b)/2$, where a is the tumour height in mm and b is the largest basal tumour diameter in mm. This was published by Richtig et al. [14] who concluded that application of this formula to choroidal melanomas was a better indicator of tumour survival than assessment of the height or basal diameter alone.

The tumour volume was calculated for each tumour at each visit where ultrasound was performed until the patient's discharge from the service, death, or enucleation of the eye. Analysis points were as follows: endpoint 1 was taken at the scan at least 6 months post-PBT; endpoint 2 was the scan at least 12 months post-PBT; and endpoint 3 was the scan at least 24 months post-PBT. As the study is retrospective, patients' appointment times could not be prospectively scheduled hence most patients were not seen exactly at 6, 12, or 24 months. The visits closest to but after the 6, 12, or 24 months timepoint were chosen so, for example, if a patient had a scan at 5 and 8 months, the 8-month scan would be the one chosen for analysis for endpoint 1. The percentage change in tumour volume from pre-PBT baseline and the months that had elapsed since treatment were also calculated for each visit. Means, medians, proportions, and ranges of values were calculated using Microsoft Excel (Microsoft, Redmond, WA, USA) and a comparison of means using Medcalc for Windows (Medcalc, Ostend, Belgium) with p values of <0.05 was taken to be statistically significant. As the study was retrospective and involved anonymised patient data, individual patient consent was not deemed to be required. The study adhered to the tenets of the Declaration of Helsinki.

Table 1. Patient characteristics

Sex, %	
Male	57.5
Female	42.5
Mean age, years	61.9
Tumour type	
Choroidal melanoma	169
Ciliary body melanoma	24
Iris melanoma	0
Laterality	
Right	91
Left	102
Previous treatment	
None	132
Ruthenium plaque	46
PDT	2
Diode laser	2
Plaque+Diode	10
Resection and plaque	1
Local recurrence after PBT	
Yes	38
No	153
Suspected	1
Metastases	
Yes	49
No	143
Uncertain	1

PDT, photodynamic therapy; PBT, proton beam therapy.

Results

195 patients referred for proton beam treatment between April 2008 and April 2018 were identified. Of these, 2 patients were excluded as they had undergone treatment for a diagnosis other than choroidal, ciliary body, or iris melanoma leaving 193 patients for analysis. Break-down of patient demographics and baseline characteristics are found in Table 1. The majority of patients were treated for choroidal melanomas and 61/193 (31.6%) of patients had some form of treatment for their tumour prior to PBT.

On pre-PBT ultrasound measurements, median height was 6.1 mm (range 0.9–15.4 mm) and a median volume of 411.5 mm³ (range 14.7–2481.7 mm³). Thirty-two patients had ultrasound scans performed between 1 and 3 months following PBT. Analysis of this subgroup showed 21 of these had a decrease in tumour volume with a mean of 16.8%, whilst 11 showed an increase compared to measurements taken prior to PBT by a mean of 71.5%. A reduction in tumour size was seen in 8 patients less than 2 months after completing PBT.

Figure 1 provides a schematic representation of the outcomes of all patients included in the study between endpoint 1 and endpoint 3. Figure 2 provides an illustrative example of a tumour which responded to PBT at endpoint 1.

At endpoint 1, a mean of 8 months post-PBT, 166 patients had ultrasounds available for analysis. Of these, 2 (1.2%) patients had a flat tumour on ultrasound so a volume could not be calculated. 125 (75.3%) patients had a decrease in tumour volume with a mean decrease of 35.2% and 39 (23.5%) patients had an increase in tumour volume with a mean of 38.4% increase.

Concentrating on patients with an increase in tumour volume at endpoint 1 ($n = 39$), by their next scheduled visit, 4 patients had an enucleation for progressive disease and one due to a painful eye, 1 patient died from metastatic disease and 1 from other causes and 2 patients did not attend for follow-up. 16 of these tumours showed a decrease from pre-treatment volume at their next visit, 8 showed a decrease from their volume at endpoint 1 but remained larger than baseline at their next visit, and 6 continued to enlarge.

At endpoint 2, a mean of 15 months post-PBT, 153 patients had ultrasounds available for analysis. 2 tumours (1.3%) remained flat, 131 patients (85.6%) had a reduction in tumour volume from baseline by a mean of 43.9% and 20 (13.1%) of patients had tumours which were larger than baseline by a mean of 37.4%.

Concentrating on those patients with an increase in tumour volume at endpoint 2 ($n = 20$), 2 patients died, 2 patients had an enucleation for progressive disease, 9 tumours shrunk to become smaller than pre-treatment at their next visit, 3 tumours decreased in size to become smaller but remained larger than pre-treatment, and 4 continued to increase in size.

At endpoint 3, a mean of 28 months post-PBT, 126 patients were available for analysis. 4 patients (3.2%) had flat tumours on ultrasound, 113 patients (89.7%) had a reduction in tumour volume compared to baseline, and 9 patients (7.1%) had tumours which remained larger than baseline.

Looking at response in patients with more than 30 months of ultrasound follow-up ($n = 114$), 1 patient took 32 months following PBT for their tumour to decrease in volume to below pre-PBT baseline with 4 taking more than 24 months to see a radiological response.

A post hoc subgroup analysis was performed on patients who had treatment for their melanoma prior to

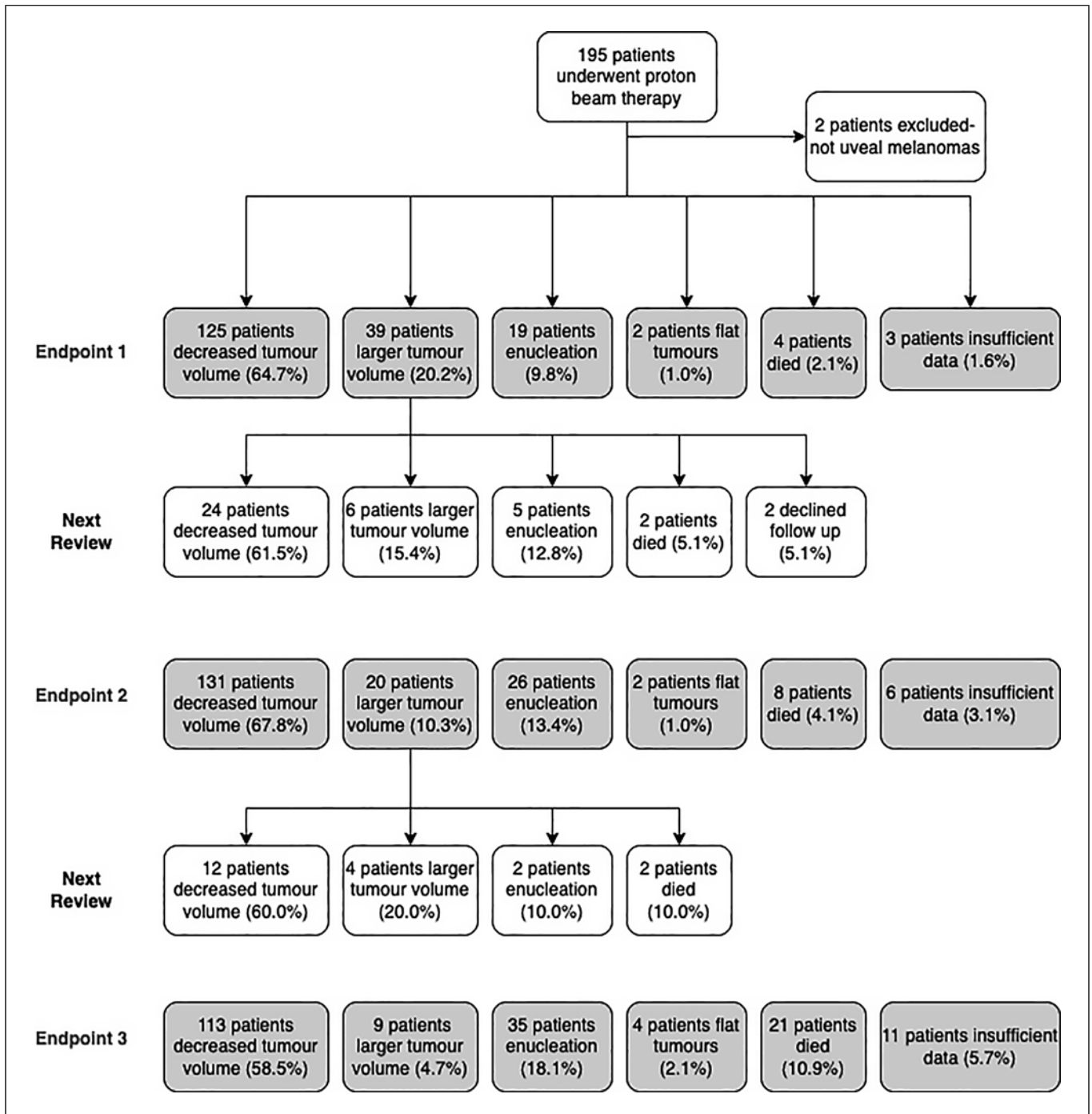


Fig. 1. Flow chart indicating outcomes for patients at endpoints 1, 2, and 3.

PBT ($n = 61$). The outcomes of this are displayed in Figure 3. The mean percentage tumour size decrease in those tumours which decreased at endpoint 1, 2, and 3 were not statistically significant when compared to the whole group ($p = 0.94$, $p = 0.82$, $p = 0.46$, respectively,

comparison of means). There was also no significant difference between the proportion of tumours decreasing or increasing in the previously treated subgroup compared to the cohort as a whole ($p = 0.25$ – 0.89 , comparison of proportion). Tumours which

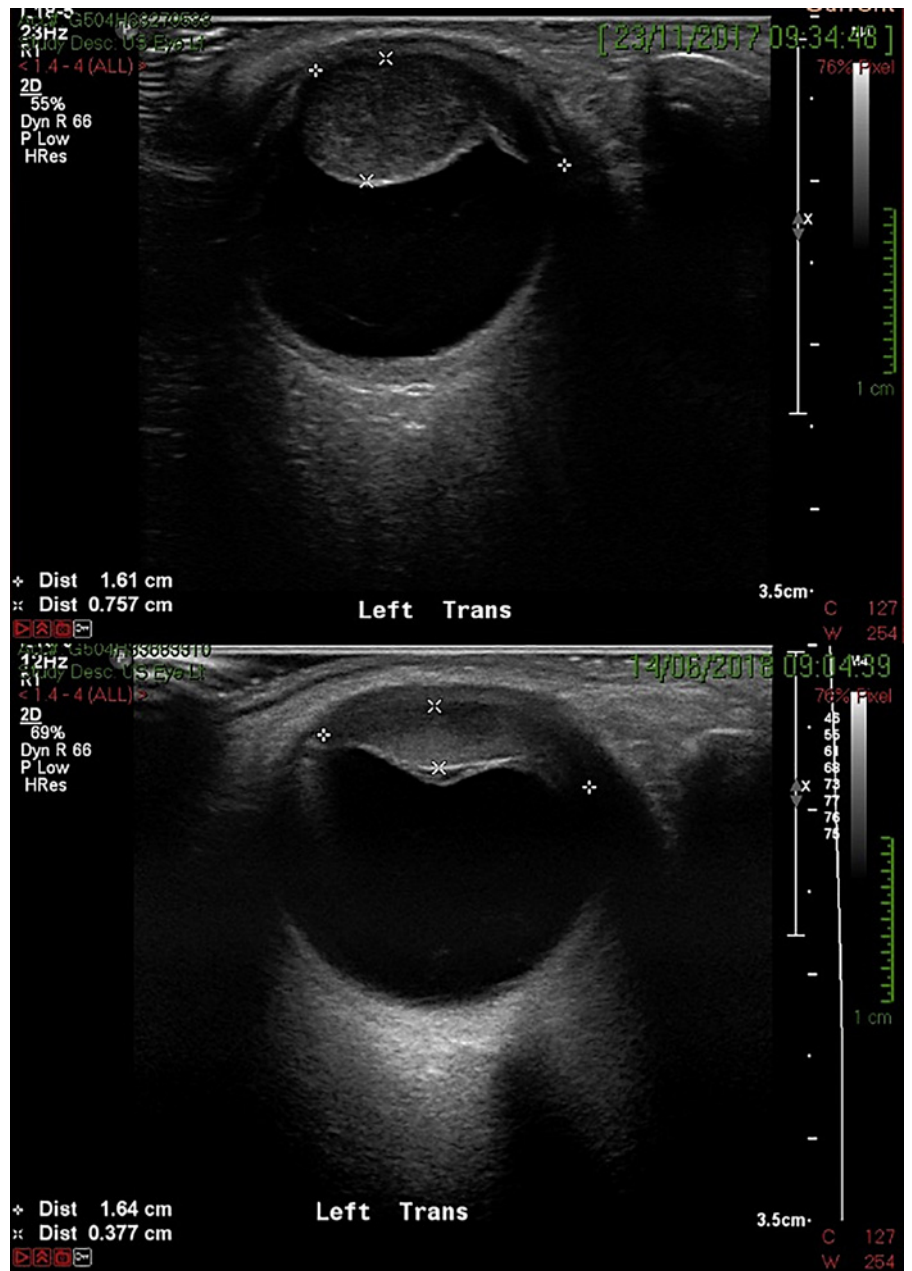


Fig. 2. Ultrasound images before PBT (top) and 6 months after PBT (bottom).

were previously treated were, however, significantly smaller pre-PBT than the group as a whole with a mean of 212 mm³ versus 585 mm³ ($p \leq 0.0001$, comparison of means).

Fifty-six patients (29%) died within 5 years of melanoma diagnosis in this study; in 38 of these (19.2% of total cases) the cause of death was melanoma related. Choroidal melanoma patients had a 5-year melanoma-specific survival of 82.3% and for ciliary body melanoma patients, this was 80.1%. Cause of death in 7 pa-

tients could not be identified and the cause of death in the remaining 11 patients was not felt to be related to their uveal melanoma.

Discussion

Our study shows that following PBT for uveal melanomas, the majority of patients have a radiological reduction in volume at around 6 months following

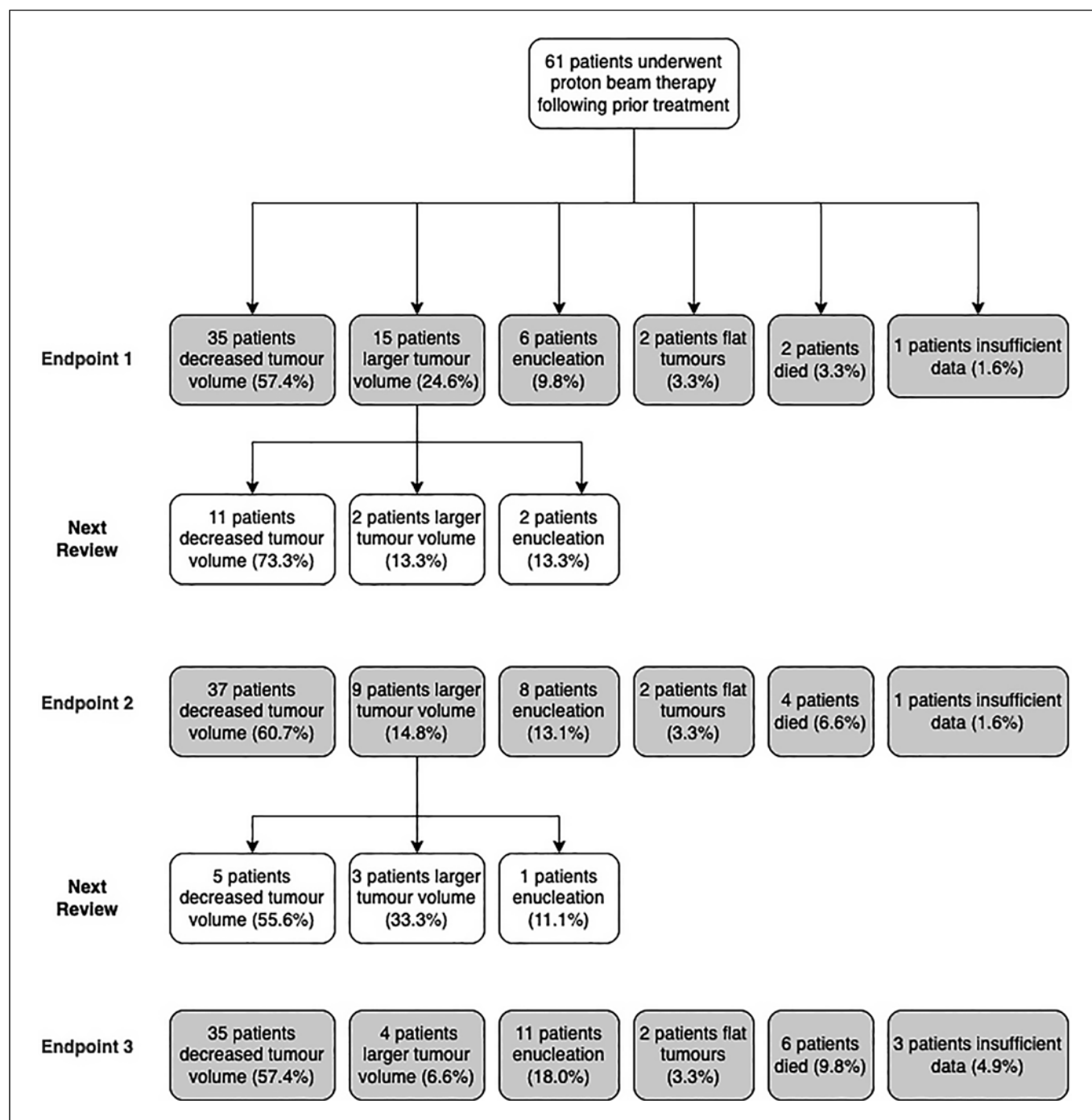


Fig. 3. Flow chart indicating outcomes for patients at endpoints 1, 2, and 3 for patients with treatment prior to PBT.

treatment. However, some patients show a radiological response as early as 2 months and others may take 24 months or more for a radiological response to be seen. We believe this is the largest series of patients looking at radiological response to PBT in uveal melanomas.

In 1982, Wilkes and Gragoudas [13] looked at a small series of uveal melanomas following proton beam irradiation. They found that when looking at changes in height in response to PBT on ultrasound, the earliest response was seen at 1 month, the average response was 12 months, and the latest response was seen at 24 months which are

consistent with our findings. They also noted that small tumours decreased more slowly than larger tumours though this was not statistically significant [2]. Foti et al. [15] used MRI to look at volume changes of uveal melanomas in response to PBT and found a significant decrease in volume on MRI at 6 months post-PBT in 12 patients, as was observed in the majority of our patients. The mean tumour size was, however, smaller than the mean tumour size in our study (387.0 mm^3 vs. 591.6 mm^3). Kim et al. [16] looked at a small series of treatment naïve patients undergoing PBT for choroidal melanoma and measured volume changes in tumours on MRI following treatment. They found an average volume reduction of 32.8% at 6 months and 39.6% at 12 months in the 23 patients with volume reduction which are comparable to our figures of 35.2% and 43.9% at around 6 and 12 months, respectively.

It is recognised that patients receiving PBT for uveal melanomas can have an increase in the size of the tumour in the first months following PBT [17]. In the 32 patients in our series who had an ultrasound at or before 3 months post-treatment, 10 (31%) patients showed an increase in tumour volume. However, at the next ultrasound scan 7 out of 10 (70%) of these were smaller so, in the absence of other features of non-response, we suggest closely monitoring these patients rather than enucleating early.

Looking specifically at patients who had melanoma treatment prior to PBT, these seem to respond similarly to PBT when compared to treatment naïve tumours in terms of the proportion of tumours responding and the proportion of response. We are not aware of any existing literature specifically looking at reductions of melanoma volume after PBT in treated and treatment naïve tumours. Reichardt et al. [18] looked at 48 previously treated uveal melanomas salvaged with PBT. These had a mean volume prior to PBT of 172.5 mm^3 so similar to our previously treated cohort which had a median of 166 mm^3 , but they did not report on the effect of PBT on tumour volume.

Jamieson et al. [3] looked at survival of all uveal melanoma patients in Scotland between 1998 and 2002 and found that the 5-year melanoma-specific survival was 92.3% for choroidal melanomas and 87.2% for ciliary body melanomas, higher than our survival of 82.3% for choroidal and 80.4% for ciliary body melanomas treated with PBT. This difference is statistically significant for choroidal but not ciliary body melanomas. We propose that this may be related to the use of PBT for choroidal melanomas which are too large for plaque treatment or have not responded to plaque treatment which would confer a survival disadvantage compared to a wider subset of choroidal melanomas.

We consider strengths of our study to be a high number of patients with at least 2 years of follow-up, and we feel that

we are likely to include all patients meeting the inclusion criteria in Scotland. Limitations of our study include its retrospective nature and that we did not refer any patients with an iris melanoma for PBT so our findings cannot be applied to this subset of uveal melanoma. Whilst there is bound to be a degree of variation between ultrasound scans, we believe the same experienced radiologists performing the scans on each occasion and scans being performed on the same scanner minimises such variation. All patients received their treatment at the same PBT location, so we believe that variation in treatment is minimal though we cannot exclude variation in these parameters affecting our results. Cytogenetic testing was less prevalent at the start of our study period and we only have cytogenetic data from patients with a tissue sample, usually from enucleation. We therefore do not have cytogenetic data for a large number of patients. We could therefore not assess the effect of cytogenetics on response of melanomas to PBT in this cohort but would recommend that this be included for any prospective studies in this area or with more recent cohorts.

In conclusion, we suggest that in the absence of other signs of progression, it is sensible to wait for 6 months post-PBT prior to considering further intervention as the majority of patients will have shown radiological response at this point. In a subset of patients, it may be appropriate to wait up to 24 months from treatment as a response could be seen only at this time. However, ultrasound evaluation is only one aspect of assessment for uveal melanoma progression or recurrence, and so clinicians must assess for other signs of progression before electing to monitor those without a clear radiological response to PBT.

Acknowledgments

The authors would like to acknowledge the work of Susan Ewan, ocular oncology coordinator in identifying patients included in the study and the team at the Clatterbridge Cancer Centre NHS Foundation Trust who undertook the PBT for all patients.

Statement of Ethics

Ethical approval and consent to participate are not required for this study in accordance with national guidelines.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

Funding Sources

The authors received no funding for the production of this work.

Author Contributions

P.C., V.C., J.C., G.A.F., O.C., and M.G. developed the idea and methodology for the work. M.G., G.A.F., O.C., J.N., O.Cr. and W.K. collected data for the manuscript. M.G. and G.A.F. drafted the manuscript which was critically appraised by all authors.

Data Availability Statement

Data used in this manuscript contains potentially identifiable information, however, is available on reasonable request. Enquiries regarding the manuscript data can be directed to the corresponding author.

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